

Sanghyun Joshua Cho

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West Lafayette, IN 47906

Systems-oriented Computer Science student with hands-on experience in compiler construction, low-level Unix systems programming, and an interest in designing and building full-stack, scalable software applications

EDUCATION

Purdue University, College of Science

West Lafayette, IN

Bachelor of Science in Computer Science, Focus: Systems Engineering

May 2027

- **Relevant Coursework:** Object-Oriented Programming in Java, Discrete Mathematics, Programming in C, Computer Architecture, Data Structures and Algorithms, Advanced Algorithms, Compilers

EXPERIENCE

Electrical Engineering Systems Assistant

Oct 2025 – Present

Purdue University

West Lafayette, IN

- Developed Python scripts to automate facility data collection and generate structured reports from sensor records, supporting data-driven engineering decisions.
- Analyzed operational datasets to guide predictive maintenance strategies across multiple research labs, contributing to sustained system uptime and improved reliability.
- Streamlined internal workflows by scripting routine processes and documenting software procedures, helping maintain **99%+** system availability across the facility.

Oathbound Studios

May 2025 – October 2025

Systems Programmer

West Lafayette, IN

- Implemented core gameplay systems using object-oriented design principles, contributing to a modular, maintainable, and memory-efficient architecture.
- Served as the primary point of contact for **C++**-related questions, providing guidance on best practices, object-oriented design, and debugging strategies to team members.
- Optimized pre-existing gameplay systems through targeted refactoring, improved memory usage, and runtime performance tuning, enhancing overall efficiency by **16%**.

PROJECTS

Beatproof

October 2025

- Lead the **Next.js + TypeScript** architecture for Beatproof's Solana audio marketplace, pairing server-rendered React with Framer Motion animation and responsive UI engineering to improve first meaningful paint by **28%**.
- Architected Phantom wallet onboarding with Solana's propriety wallet along with side secure Next.js session APIs, reducing wallet drop and improving successful marketplace entries during internal QA.
- Drove deep stack integration—optimizing Solana RPC calls with Anchor utilities, unifying **Zod** validation, and modernizing bundling/linting—which shortened regression test cycles by **45%** and stabilized quasi-production QA environments.

Heater App – iOS

Aug 2025 – Oct 2025

- Developed and launched a SwiftUI iOS app for logging and reviewing sports games using an **MVVM architecture** with modular Views, Models, and Services.
- Integrated **Firebase Auth** and Firestore for secure user authentication, real-time data persistence, and seamless cross-device synchronization.
- Deployed to a college community of over **750 users**, with a local server API for sports metadata (teams, schedules, stats) and caching that reduced redundant network requests by **21%**, improving responsiveness and load times.

Unix Shell

Jul 2025

- Built a Unix-like shell in C++ and C supporting process creation, built-in commands, file redirection, and pipelines.
- Implemented background job control using SIGINT and SIGCHLD to handle concurrent multi-process execution.
- Designed a modular parsing and execution pipeline for complex commands and multi-stage I/O operations.

SKILLS

Languages: C++, C, Swift, Python, Bash, JavaScript

Technologies: Firebase, Unreal Engine, Lex/Yacc, LLVM, Git, Version Control, Unix/Linux Systems, Next.js, Phantom

Concepts: Compiler Design, Systems Programming, Memory Management, Object-Oriented Programming (OOP), Data Structures & Algorithms